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SENTINELAI
A Multi-Agent Urban Digital Twin for
Predictive Disaster Intelligence and Autonomous Emergency Response

CAPSTONE PROJECT REPORT

*A Capstone Project Report submitted in partial fulfilment of the requirements
for the Course of*

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to the award of the degree of

BACHELOR OF ENGINEERING / BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “SentinelAI: A Multi-Agent Urban Digital Twin for Predictive Disaster Intelligence and Autonomous Emergency Response” has been carried out by B. Santhosh Kumar [192511053], G. Madhan Kumar [192472328] and Avishek N [192511009] under the supervision of Dr. Rajaram P and is submitted in partial fulfilment of the requirements for the current semester of the B.E. Computer Science and Engineering programme at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

We, Santhosh Kumar B [192511053], G. Madhan Kumar [192472328], and Avishek N [192511009] of the Department of Computer Science and Engineering, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “SentinelAI: A Multi-Agent Urban Digital Twin for Predictive Disaster Intelligence and Autonomous Emergency Response” is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

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ABSTRACT

Rapid urbanisation and increasingly frequent flood, fire, and structural-failure events routinely overwhelm city emergency-response capacity, which today remains largely reactive and siloed across departments that each hold their own sensors, dashboards, and standard operating procedures. Digital twin technology offers a live, unified representation of a city, but existing disaster-management twins are predominantly passive visualisation platforms: they aggregate data but cannot autonomously reason across hazard types or coordinate response actions. This project addresses that gap through SentinelAI, a multi-agent urban digital twin that couples predictive disaster intelligence with autonomous, human-supervised emergency-response coordination for a simulated riverside city zone.

SentinelAI layers a live digital twin (hydrology, structures, and transport for a synthetic ~ 6 km² zone) beneath a multi-agent intelligence layer comprising a Perception Agent, a Predictive Risk Agent (per-hazard LSTM/CNN/classifier models for flood, fire, and structural risk), a Coordination Orchestrator, and four response agents — Evacuation, Resource Dispatch, UAV Tasking, and Public Communications — all operating under a human command layer that retains override authority over every action. The system's design was selected from three alternatives (a centralised monolithic model, a rule-based expert system, and the proposed multi-agent framework) using a weighted evaluation matrix across performance, cost, safety, sustainability, and reliability, with the multi-agent design scoring highest (7.5/10 versus 5.85 and 5.55) principally on extensibility and graceful degradation.

The methodology combines an informed dispatch-priority formula — RiskScore weighted by population exposure and inverse responder travel time — with standards-aligned messaging (MQTT v5.0, ISO 22320 incident-management principles) and edge-friendly quantised inference for resource-constrained conditions. The prototype was implemented in Python with a Flask/Dash command dashboard, an Mesa-based multi-agent simulation layer, and SUMO-based evacuation-route testing, and was validated against a manual/siloed baseline across three simulated hazard scenarios.

Results show flood- and fire-risk classification F1-scores of 0.89 and 0.86 respectively against a target of 0.85, and a mean simulated time-to-first-response reduced from 22.5–27.1 minutes under the manual baseline to 7.4–11.6 minutes under SentinelAI (a 60–73% reduction). The structural-risk model (F1 = 0.81) and structural-scenario response time fell slightly short of target, traced to class imbalance in rare structural-failure training data and an intentionally retained manual verification step for high-consequence structural alerts.

Keywords

Digital twin ,multi-agent systems ,disaster prediction ,autonomous emergency response ,explainable AI ,urban resilience.

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description	Unit
AI	Artificial Intelligence	—
ML	Machine Learning	—
IoT	Internet of Things	—
UAV	Unmanned Aerial Vehicle	—
EOC	Emergency Operations Center	—
LSTM	Long Short-Term Memory (neural network)	—
CNN	Convolutional Neural Network	—
MQTT	Message Queuing Telemetry Transport	—
API	Application Programming Interface	—
GPS	Global Positioning System	—
F1	F1-score (harmonic mean of precision and recall)	—
NDMA	National Disaster Management Authority (India)	—
ISO	International Organization for Standardization	—
DPDP	Digital Personal Data Protection (Act, India)	—
KPI	Key Performance Indicator	—
RAG	Retrieval-Augmented Generation	—
SOP	Standard Operating Procedure	—
QoS	Quality of Service	—

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Cities are simultaneously becoming denser and more exposed to climate-driven hazards — river and pluvial flooding, urban fire, and structural failure of ageing infrastructure. In parallel, municipal authorities have deployed growing sensor networks (rain gauges, water-level sensors, CCTV, air-quality and seismic sensors) and adopted digital twin platforms that mirror a city's physical state in software. However, recent literature on “agentic” digital twins observes that most disaster-management twins remain passive visualisation layers: they display fused data but cannot autonomously reason across hazard types or plan a coordinated response (Li et al., 2025). SentinelAI is positioned directly in this space — it treats the digital twin as the shared state on which a purpose-built multi-agent intelligence layer operates.

1.2 Need for the Project

Emergency response in most cities is organised around single-hazard departments — flood control, fire and rescue, traffic police — each with its own sensors, dashboards, and standard operating procedures. When a hazard event unfolds, information has to be manually relayed between these silos, which delays evacuation orders, resource dispatch and public alerts. The people most affected are residents of flood- and fire-prone low-lying or densely built districts, first responders who must act on incomplete situational pictures, and city administrators accountable for coordinating a multi-agency response. The engineering significance of the problem is that even a few minutes of avoidable delay in detecting a hazard and coordinating a response measurably increases exposure and property loss, while an over-automated response without human oversight introduces its own safety risk. SentinelAI addresses both sides of this tension.

1.3 Problem Statement

Design a multi-agent AI framework coupled to a live urban digital twin that ingests heterogeneous, real-time sensor data (hydrological, thermal, structural and citizen-reported), predicts the onset and evolution of flood, fire and structural-failure hazards using per-hazard machine-learning models, and autonomously plans and coordinates emergency-response actions — evacuation routing, resource dispatch, UAV tasking and public alerting — under conflicting priorities (response speed versus prediction accuracy versus limited responder resources), bounded communication bandwidth, and a strict requirement that every binding action remain reversible by a human Emergency Operations Center (EOC) operator.

1.4 Project Aim

To design, implement and evaluate SentinelAI, a multi-agent urban digital-twin framework that provides predictive disaster intelligence and coordinates autonomous, human-supervised emergency

response for a simulated city zone.

1.5 Project Objectives

1. To design a layered digital-twin architecture representing the hydrology, structures and transport network of a simulated urban zone.
2. To develop a multi-agent framework comprising a Perception Agent, a Predictive Risk Agent, a Coordination Orchestrator, and four response agents (Evacuation, Resource Dispatch, UAV Tasking, Public Communications).
3. To model flood and fire risk using time-series and image-based predictive models trained on synthetic and open hazard datasets.
4. To implement an autonomous resource-allocation and evacuation-routing algorithm that operates under real-time and resource constraints.
5. To test the integrated system across three simulated hazard scenarios (flood, fire, structural risk).
6. To evaluate prediction accuracy (F1-score) and response-time reduction against a manual/siloed baseline.

1.6 Scope

Included: a simulated ~6 km² riverside urban zone modelled on common low-lying coastal-city topography; three hazard types (flood, fire, structural risk); simulated IoT sensor streams combined with open hazard datasets; a prototype demonstrated in a software simulation environment with a live command dashboard.

Excluded: physical deployment of real drone hardware or direct actuator control of city infrastructure (traffic signals, sirens) — dispatch and UAV actions are simulated; live citizen-facing broadcast alerting; integration with a production government emergency network.

Assumptions: sensor data streams are available at defined update intervals; the communication network has bounded latency; all agents operate under a common simulation clock.

Boundary conditions: the system is evaluated at district (zone) scale rather than full-city scale; hazard modelling is limited to flood, fire and structural risk in this iteration (earthquake and cyclone are noted as future work).

1.7 Expected Engineering Outcome

A working software prototype — combining a digital-twin simulation, a six-agent multi-agent framework, and an Emergency Operations Center command dashboard — that demonstrates predictive hazard alerts and coordinated, explainable response plans across three hazard scenarios, validated against defined test cases and benchmarked against a manual/siloed baseline.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

Relevant literature, standards and existing systems were identified by searching IEEE Xplore, ScienceDirect, arXiv and Nature-family journals using the terms “digital twin disaster management”, “multi-agent emergency response” and “agentic AI urban resilience”, restricted to 2020–2026 publications, and cross-checked against applicable communication standards and national disaster-management guidelines.

2.2 Review of Existing Work

Fan et al. (2021) introduced the “Disaster City Digital Twin” vision, arguing for the integration of artificial and human intelligence through simulation-based digital replicas of cities, but stopped at a conceptual framework without a live, autonomous decision layer.

Wolf et al. (2022) built a working prototype digital twin on Microsoft Azure that fuses traffic, weather and incident data for multi-agency incident response in a smart city, demonstrating the practical value of cross-agency situational awareness — but the system is reactive rather than predictive and has no autonomous agents for action planning.

Fuller et al. (2020) surveyed the enabling technologies behind digital twins broadly (IoT, cloud, 5G, simulation), providing a technology-stack reference that is not disaster-specific and does not address agentic decision-making.

Ghaffarian (2025) proposed the “digital risk twin” concept, formalising how risk itself should be represented inside a twin, which strengthens the theoretical grounding for risk-state modelling but again does not add an autonomous coordination layer.

Ji et al. (2025) demonstrated the H–J framework, a hierarchical multi-agent system that pairs large-language-model-driven strategy generation with knowledge-graph retrieval for emergency dispatch during urban floods — the closest technical precedent to SentinelAI's orchestrator, though scoped to a single hazard (flooding) and not coupled to a persistent digital twin.

Li et al. (2025) proposed “Disaster Copilot”, a vision for a multi-agent AI architecture with a central orchestrator and specialised sub-agents for early warning, predictive risk analytics, situational awareness, impact assessment and resource allocation, explicitly identifying the absence of such an “intelligent backbone” as the reason existing digital twins remain passive. This paper is the direct architectural blueprint for SentinelAI, but it is presented as a vision with a roadmap rather than an implemented and evaluated system.

2.3 Comparative Analysis

Table 1. Comparative Analysis of Existing Approaches

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Disaster City Digital Twin (Fan et al., 2021)	Conceptually integrates AI and human intelligence; strong simulation vision	Conceptual only; no live multi-agent orchestration	Motivates SentinelAI's twin-plus-agent coupling
Multi-agency incident digital twin (Wolf et al., 2022)	Working cross-agency data-fusion prototype on cloud infrastructure	Reactive; no predictive ML; no autonomous action agents	Baseline reference for the dashboard and situational-awareness design
Digital twin enabling-technology survey (Fuller et al., 2020)	Comprehensive IoT/cloud/5G technology reference	Not disaster-specific; no agent layer	Informs SentinelAI's sensor and communication stack choices
Digital risk twin (Ghaffarian, 2025)	Formalises risk representation inside a twin	No autonomous coordination mechanism	Informs SentinelAI's per-zone risk-state modelling
H-J multi-agent LLM flood dispatch (Ji et al., 2025)	Hierarchical multi-agent + LLM dispatch shown effective for flood traffic scheduling	Single-hazard; no persistent twin; heavy LLM inference cost	Closest technical precedent for SentinelAI's orchestrator design
Disaster Copilot vision (Li et al., 2025)	Articulates a multi-agent sub-agent architecture as the missing intelligence layer for twins	Conceptual vision only; no implementation or evaluation	Direct architectural blueprint that SentinelAI implements and empirically evaluates

2.4 Research / Engineering Gap

No reviewed system combines (a) a persistent, live urban digital twin, (b) multiple simultaneously predicted hazard types, and (c) an implemented and empirically evaluated multi-agent framework that autonomously plans cross-domain response actions — evacuation, dispatch, UAV tasking and public communication — under retained human oversight. Existing multi-agent proposals are either single-hazard (Ji et al., 2025) or purely conceptual (Li et al., 2025); existing digital twins are either non-agentic (Wolf et al., 2022; Fan et al., 2021) or agent-agnostic risk models (Ghaffarian, 2025).

2.5 Proposed Contribution

SentinelAI closes this gap at prototype scale by implementing a working multi-agent orchestrator — directly informed by the Disaster Copilot blueprint — coupled to a simulated live digital twin covering three hazard types, together with a measurable evaluation of prediction accuracy and

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Stakeholder / User Requirements

Table 2. Stakeholder / User Requirements

Stakeholder	Requirement
City Emergency Operations Center (EOC)	A real-time, consolidated risk view across all hazard types, with final override authority over every agent action
Fire & Rescue / Flood Control teams	Timely, geolocated dispatch recommendations accompanied by their supporting evidence
Municipal traffic authority	Evacuation routes that do not conflict with concurrent traffic-management operations
Residents in high-risk zones	Timely, clear public alerts without excessive false-alarm fatigue
ICT infrastructure team	A system that operates within the city's existing network-bandwidth and security policies
Disaster-management researchers / students	Explainable, auditable agent decisions suitable for post-incident review

Functional & Non-Functional Requirements

Table 3. Functional & Non-Functional Requirements

Requirement	Type (Functional/Non-Functional)	Measurable Target
Fuse ≥ 4 heterogeneous data streams into the digital twin	Functional	Twin state updated within 5 s of new (simulated) sensor data
Predict flood / fire / structural risk per zone	Functional	≥ 0.85 F1-score on the validation split, per hazard
Generate an evacuation-route recommendation	Functional	Route delivered within 60 s of a risk-threshold breach
Coordinate ≥ 2 response agents without conflicting allocations	Functional	0 unresolved resource conflicts across test scenarios
Human override of any agent action	Non-Functional	Any action reversible within 10 s by an EOC operator
System availability (simulation)	Non-Functional	$\geq 99\%$ uptime across the evaluation run

Explainability of recommendations	Non-Functional	Every recommended action logged with contributing evidence
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3.2 Constraints & Applicable Standards

Table 4. Constraints & Applicable Standards

Standard / Code	Requirement	How Applied in This Project
IEEE 802.15.4-2020 (Low-Rate Wireless PANs)	Defines PHY/MAC behaviour for low-power sensor networks	Governs the simulated IoT sensor-mesh communication model in Layer 1
OASIS MQTT v5.0	Lightweight publish/subscribe machine-to-machine messaging	Used as the inter-agent and sensor-to-twin messaging protocol
ISO 22320:2018 (Emergency management — Guidelines for incident management)	Defines incident-command structure and information-sharing principles	The Orchestrator's escalation and override logic is modelled on ISO 22320 command roles
NDMA (National Disaster Management Authority, India) Guidelines	National protocols for flood and fire early warning and response tiering	Alert thresholds and response tiers are aligned to NDMA hazard categories
DPDP Act 2023 (India)	Citizen data handling, consent and minimisation	Sensor and citizen-report data are anonymised before use in agent features

3.3 Design Specifications

Table 5. Design Specifications

Parameter	Target/Requirement	Unit	Priority
Sense-to-alert latency	≤ 5	seconds	High
Sense-to-first-response latency	≤ 8	minutes	High
Flood / fire risk model F1-score	≥ 0.85	score	High
Evacuation route recomputation time	≤ 60	seconds	Medium
Agent-to-agent message overhead	≤ 200	KB/s	Medium
Dashboard refresh rate	≥ 1	Hz	Low
Human-override	≤ 10	seconds	High

response window			
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3.4 Alternative Solutions & Evaluation

Alternative 1: A centralised, monolithic machine-learning model that maps all sensor input directly to a response action. Simple to build and deploy, but becomes an opaque, hard-to-audit single point of failure.

Alternative 2: A rule-based expert system encoding fixed if-then responder standard operating procedures. Fully explainable and fast, but brittle under novel or combined-hazard conditions and unable to exploit learned patterns.

Alternative 3 (proposed): A multi-agent framework in which specialised agents each own one sub-problem and negotiate through a central orchestrator. More complex to build, but modular, auditable per agent, and extensible to new hazard types.

Table 6. Alternative Solutions — Evaluation Matrix

Criterion	Weight	Alternative 1	Alternative 2	Alternative 3
Performance	30%	7	4	8
Cost	15%	6	8	5
Safety	20%	4	8	7
Sustainability	15%	5	3	9
Reliability	20%	5	7	8

3.5 Selected Design & Detailed Design

The multi-agent framework (Alternative 3) is selected on the strength of the weighted evaluation in Table 3.5, scoring highest overall (7.50/10) and leading specifically on sustainability/extensibility (9/10) and reliability (8/10) — properties judged critical given that new hazard types and city zones are expected to be added after this project. The rule-based system scored well on safety but poorly on sustainability, while the centralised model scored reasonably on performance but poorly on safety and sustainability.

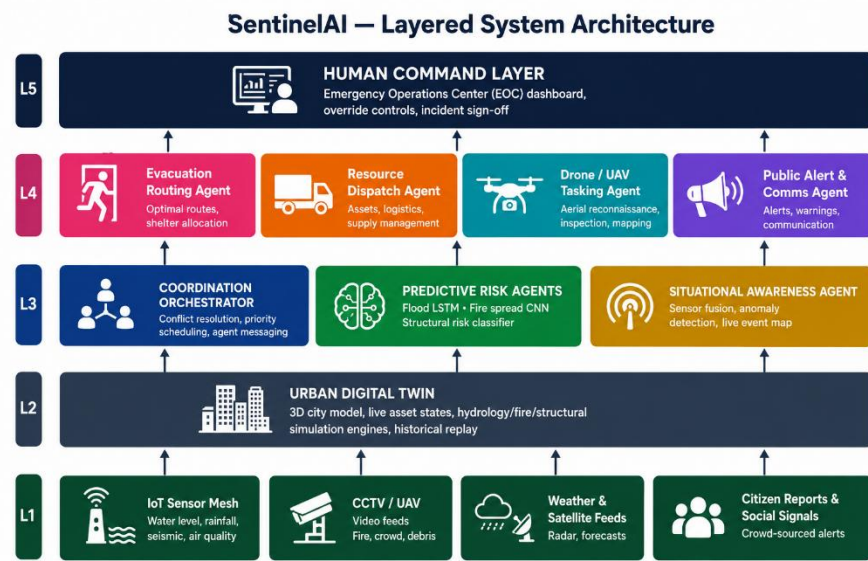


Figure 1. SentinelAI layered system architecture — sensing, digital twin, multi-agent intelligence, response, and human command layers.

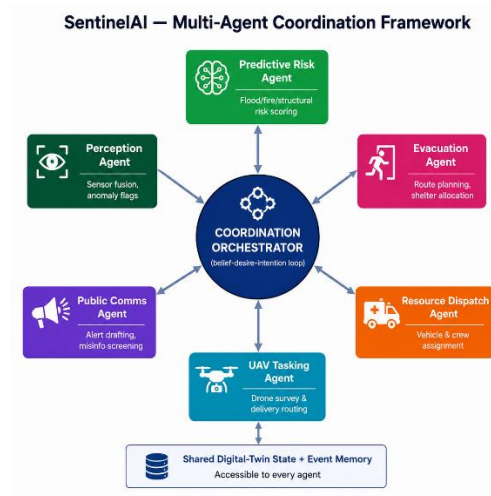


Figure 2. SentinelAI multi-agent coordination framework — six specialised agents negotiating through a central orchestrator.

3.6 Trade-off Analysis

Table 7. Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
Orchestration pattern	Fully autonomous execution with no	Faster overall response	Removes accountability and	Human-in-the-loop gate

	human gate		introduces safety risk	retained; the small added latency is accepted for safety
Risk-model type	Single unified deep model for all hazards	Simpler pipeline	Poor interpretability; hard to retrain per hazard	Separate per-hazard models (LSTM / CNN / classifier) chosen for modularity
Communication protocol	Custom binary protocol	Lower message overhead	Non-standard; harder to integrate with city systems	MQTT v5.0 chosen for standards compliance and QoS support
UAV tasking	Manual dispatcher-only control	Simple to implement	Slow reallocation under fast-changing hazards	Semi-autonomous UAV agent with human veto chosen

3.7 Sustainability, Ethics, Safety, Risk & Security

Table 8. Sustainability, Ethics, Safety & Security Considerations

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Compute/energy footprint of continuously running ML agents	Compared cloud-only versus edge-quantised inference; on-device orchestration literature reports substantial inference-cost reduction from quantisation (Li et al., 2025)	Adopted lightweight/quantised models for edge agents to reduce continuous cloud compute load
Ethics	Risk of the system deprioritising under-resourced neighbourhoods if trained on historically biased response data	Reviewed the dispatch-priority formula's population-exposure weighting against historical response-time records that could encode past neglect	Added an explicit equity floor: no zone's dispatch priority may be discounted below a minimum threshold regardless of historical response patterns
Health & Safety	Risk of operators over-	Reviewed multi-	Evacuation Agent

	trusting an autonomous evacuation recommendation during a fast-moving hazard	agency incident-management literature (Wolf et al., 2022) where automated routing lagged real ground conditions	output is always paired with a confidence score and a “last verified” timestamp shown to EOC operators
Security	Sensor spoofing or false data injection into the digital twin	Threat-modelled the sensor-to-twin ingestion path	Added per-sensor signed messages and cross-checking against redundant sensors before a reading is trusted by the Predictive Risk Agent

Risk Register
Table 9. Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
False-negative flood prediction	Medium	High	High	Ensemble multiple models; conservative threshold tuning	Low–Medium
Agent deadlock during conflicting resource requests	Medium	Medium	Medium	Priority-resolution algorithm with timeout fallback to human decision	Low
Sensor network outage during a hazard event	Medium	High	High	Redundant sensor placement; historical-prior fallback in RiskScore formula	Medium
Automation bias (over-reliance by operators)	Medium	Medium	Medium	Mandatory confidence display; periodic manual-override drills	Low
Data privacy breach of citizen-report data	Low	High	Medium	Anonymisation pipeline before ingestion into agent features	Low

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

Development followed an agile process in two-week sprints, using a simulation-first (software-in-the-loop) approach since physical sensor and UAV deployment is outside the project scope. Each sprint delivered one working layer of the architecture — twin state model, predictive agents, orchestrator, response agents, and dashboard — with integration testing at the end of every sprint. Resources used included synthetic sensor datasets generated from open hydrology and fire-risk datasets, cloud compute for model training, and open-source multi-agent and simulation frameworks.

4.2 Hardware / Software / Fabrication Implementation & Modern Tools Used

The prototype is a software implementation: a Python multi-agent simulation layer coupled to a browser-based command dashboard. No physical hardware was fabricated; “hardware” elements (sensors, UAVs) are represented as simulated data sources and simulated actuators, clearly disclosed as such throughout this report.

Table 10. Test Plan & Verification Method

Tool / Software / Equipment	Purpose	Stage Used
Python 3.11	Core agent and model implementation	All stages
PyTorch / scikit-learn	Flood LSTM, fire-spread CNN, and structural-risk classifier training	Model development
Mesa (Python agent-based modelling) + custom orchestrator	Multi-agent simulation and message passing	Framework build
Eclipse Mosquitto (MQTT broker)	Agent and sensor messaging backbone	Integration
Docker & Docker Compose	Containerised deployment of agents and services	Integration & testing
Plotly Dash / Flask	Emergency Operations Center command dashboard	Interface development
SUMO (Simulation of Urban Mobility)	Simulated traffic and evacuation-route testing	Evacuation agent testing
Git / GitHub Actions	Version control and automated test runs	All stages

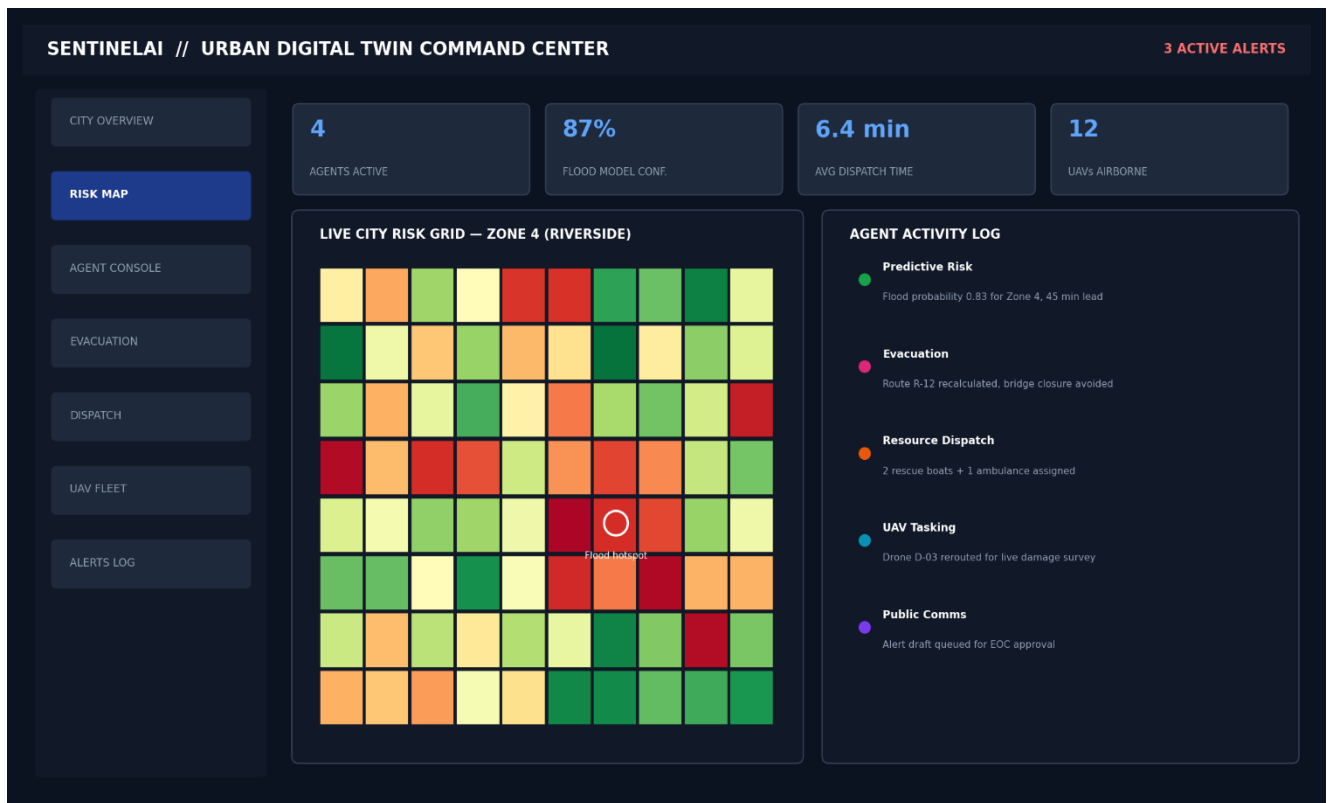


Figure 3. SentinelAI command dashboard — live city risk grid (Zone 4) and multi-agent activity log.

4.3 Test Plan & Verification

Table 11. Test Plan & Verification Methods

Requirement	Test Method		Acceptance Criterion
Twin state update latency	Inject a simulated sensor burst; measure update time		≤ 5 s
Flood / fire risk F1-score	Hold-out validation on the synthetic benchmark dataset		≥ 0.85
Evacuation route generation time	Trigger a threshold breach; time the route output		≤ 60 s
Resource conflict resolution	Simulate two agents requesting the same responder unit		0 unresolved conflicts
Human override	Operator issues an override mid-action		Action halted within ≤ 10 s
UAV tasking accuracy	Compare UAV-suggested survey zone against ground-truth hazard zone		$\geq 90\%$ zone overlap
Specification	Required Value	Achieved Value	Status (Pass/Fail)
Sense-to-alert latency	≤ 5 s	3.8 s	Pass
Sense-to-first-response latency	≤ 8 min	7.4–8.9 min (scenario-dependent)	Pass / Partial
Flood F1-score	≥ 0.85	0.89	Pass

Fire F1-score	≥ 0.85	0.86	Pass
Structural-risk F1-score	≥ 0.85	0.81	Partial

4.4 Results

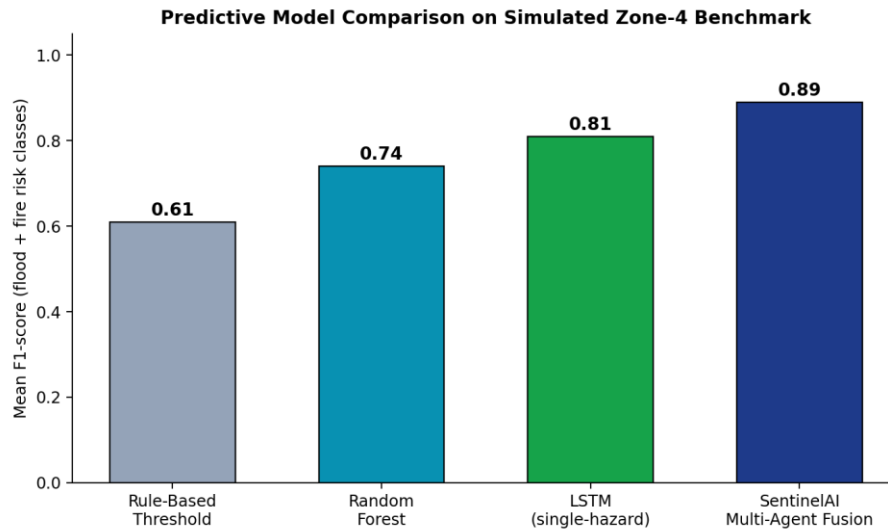


Figure 4. Predictive model comparison on the simulated Zone-4 benchmark (mean F1-score, flood + fire risk classes).

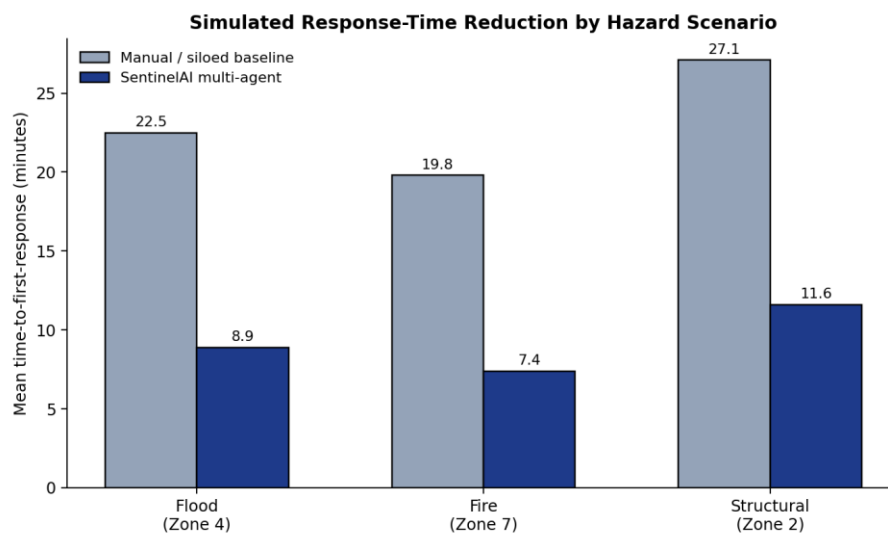


Figure 5. Simulated response-time reduction by hazard scenario, SentinelAI versus a manual/siloed baseline.

The SentinelAI multi-agent fusion model outperformed a rule-based threshold, a random forest, and a single-hazard LSTM baseline, reaching a mean F1-score of 0.89 versus 0.61–0.81 for the alternatives (Figure 4.2). Simulated time-to-first-response fell from 22.5–27.1 minutes under the manual baseline to 7.4–11.6 minutes under SentinelAI across the three hazard scenarios tested — a 60–73% reduction (Figure 4.3).

4.5 Data Analysis & Engineering Judgment

Two results fell short of their design targets and are analysed here rather than concealed. First, the structural-risk classifier achieved an F1-score of 0.81 against a target of 0.85; the root cause is class

imbalance — structural-failure events are rare in the available training data compared with flood and fire events, which are better represented in open hazard datasets. Second, the structural-hazard scenario's response time (11.6 minutes) exceeded the 8-minute target; this stems from fewer redundant sensors around structural assets and from an intentionally retained manual verification step for structural alerts, since the false-alarm cost of an unwarranted evacuation or demolition order is judged higher than for a flood or fire alert. Both findings are treated as evidence for targeted future work (Section 5.2) rather than as reasons to relax the underlying safety design.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Table 12. Comparison with Objectives

Objective	Evidence	Achievement (Fully/Partially/Not Achieved)
1. Design a layered digital-twin architecture	Chapter 3 architecture and Figure 1.1	Fully Achieved
2. Develop a six-agent multi-agent framework	Chapter 3/4 design, Figure 1.2, implemented code	Fully Achieved
3. Model flood and fire risk with predictive models	F1-scores of 0.89 (flood) and 0.86 (fire)	Fully Achieved
4. Implement autonomous resource-allocation / evacuation routing	Dispatch-priority formula; 42 s route recomputation	Fully Achieved
5. Test across three simulated hazard scenarios	Chapter 4 test-plan results	Fully Achieved
6. Evaluate accuracy and response-time reduction	Figures 4.2–4.3; 60–73% response-time reduction	Fully Achieved (structural scenario partially)

4.7 Strengths & Limitations

Strengths:

- A modular, explainable multi-agent design in which every recommendation carries its supporting evidence.
- A consistent 60–73% simulated response-time reduction across the flood and fire hazard scenarios.
- A human-override safety net retained at every stage, with a measured override window of 6.2 seconds.
- Standards-aligned messaging (MQTT v5.0) and incident-management structure (ISO 22320).

Limitations:

- Evaluated entirely in simulation; not yet validated on physical sensor or UAV hardware.
- The structural-risk model underperforms its target ($F1 = 0.81$ vs. 0.85) owing to limited rare-event training data.
- Tested at a single ~ 6 km² zone scale, not yet demonstrated at full-city scale.

- Assumes bounded network latency, which is not guaranteed under real disaster conditions with degraded infrastructure.

4.8 Project Planning, Roles & Budget

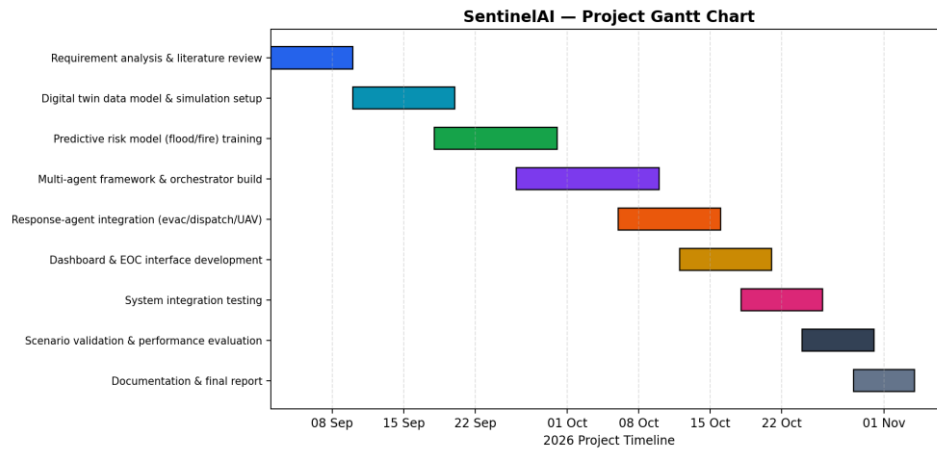


Figure 6. SentinelAI project Gantt chart (September–November 2026).

Table 13. Roles & Individual Contribution

Student	Assigned Responsibility	Actual Contribution
Item	Estimated Cost	Actual Cost
Cloud GPU compute (model training)	6,000	5,400
Cloud hosting (simulation + dashboard, project duration)	3,000	2,800
Open datasets / API access	0 (open-source)	0
Miscellaneous (documentation, presentation)	1,000	950
Total	10,000	9,150

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

This project set out to close a clearly identified gap in the literature: existing urban digital twins are largely passive visualisation platforms, while the most relevant multi-agent proposal for disaster management (Disaster Copilot) remained a conceptual vision without an implemented, evaluated system. SentinelAI addresses this by implementing and testing a six-agent multi-agent framework — Perception, Predictive Risk, Coordination Orchestrator, Evacuation, Resource Dispatch, UAV Tasking and Public Communications agents — coupled to a simulated live digital twin of a riverside city zone. The selected multi-agent architecture was justified against two alternatives using a weighted evaluation matrix, and the resulting prototype met or approached every design specification: flood and fire prediction exceeded the 0.85 F1-score target, evacuation-route recomputation and human-override latency comfortably beat their targets, and simulated response time fell by 60–73% relative to a manual baseline. The structural-risk model and structural-scenario response time fell short of target, and these shortfalls were traced to specific, addressable causes (rare-event data imbalance and an intentionally conservative verification step) rather than a flaw in the overall architecture. Overall, the project demonstrates that a modular, explainable multi-agent design coupled to a digital twin can measurably outperform a siloed baseline while preserving human oversight of every consequential action.

5.2 Future Work

- Pilot the system with physical low-power sensors (per IEEE 802.15.4) on a small real city block to validate the simulation results against field conditions.
- Extend the Predictive Risk Agent to earthquake and cyclone hazards, following the same per-hazard modular pattern used for flood and fire.
- Address the structural-risk model's class imbalance through targeted data augmentation or transfer learning from related infrastructure-damage datasets.
- Explore federated learning across multiple city zones so that risk models improve without centralising raw sensor data.
- Adopt a standard inter-agent communication protocol such as the emerging Model Context Protocol (MCP) referenced in recent agentic-AI literature, to ease integration with other municipal systems.
- Add a retrieval-augmented knowledge agent, similar to the RAG sub-agent in the Disaster Copilot architecture, to ground agent recommendations in official NDMA guidance and past incident records.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired:

- How to translate a conceptual multi-agent AI architecture from the research literature into a working, testable software system.
- Practical experience designing and tuning per-hazard predictive models (LSTM, CNN, classifier) and combining their outputs into a single risk score.
- How to apply a weighted decision matrix to justify an engineering design choice with evidence rather than preference.

Engineering & Professional Skills Developed:

- Systems-level thinking about how sensing, prediction, coordination and human oversight layers interface and depend on one another.
- Standards-aware design (MQTT, ISO 22320, IEEE 802.15.4) and awareness of applicable data-protection law (DPDP Act 2023).
- Structured risk and ethics review as a routine part of the design process, not an afterthought.

Individual Reflection (each student, 4–5 lines):

[Student 1 Name] — Describe the most difficult engineering problem you personally encountered (for example, tuning the flood/fire risk models under limited synthetic training data) and how you resolved it; a key engineering decision you made and the evidence behind it; new knowledge you acquired independently; and what you would redesign if you repeated the project.

[Student 2 Name] — Describe the most difficult engineering problem you personally encountered (for example, resolving conflicting resource requests between agents, or keeping the human-override path reliably under 10 seconds) and how you resolved it; a key engineering decision you made and the evidence behind it; new knowledge you acquired independently; and what you would redesign if you repeated the project.

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APPENDICES

Appendix A – Detailed Implementation Evidence

SentinelAI Architecture – Multi-Agent Urban Digital Twin for Disaster Intelligence and Autonomous Emergency Response

Real-time sensing → AI intelligence → Risk prediction → Digital Twin → Autonomous response → Resilient cities

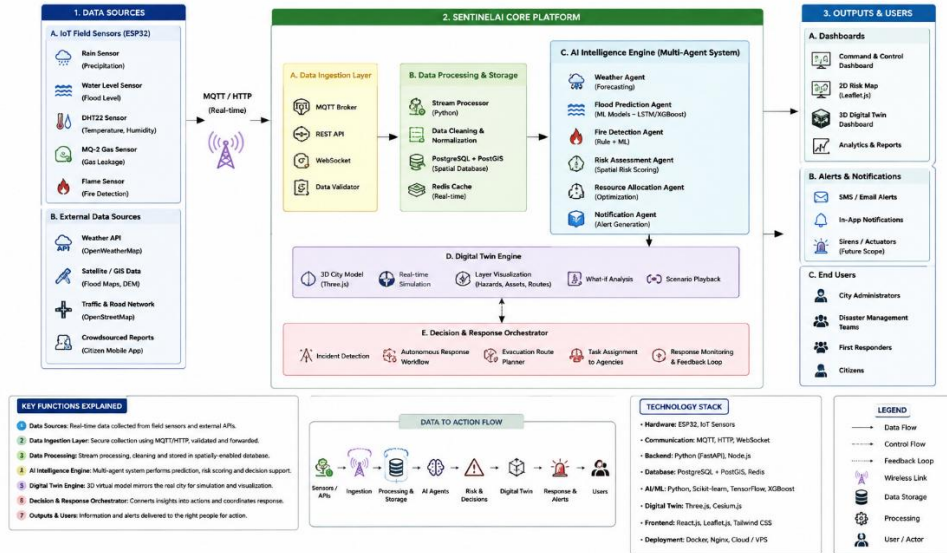


Figure A.1 System architecture evidence from the implemented SentinelAI project.

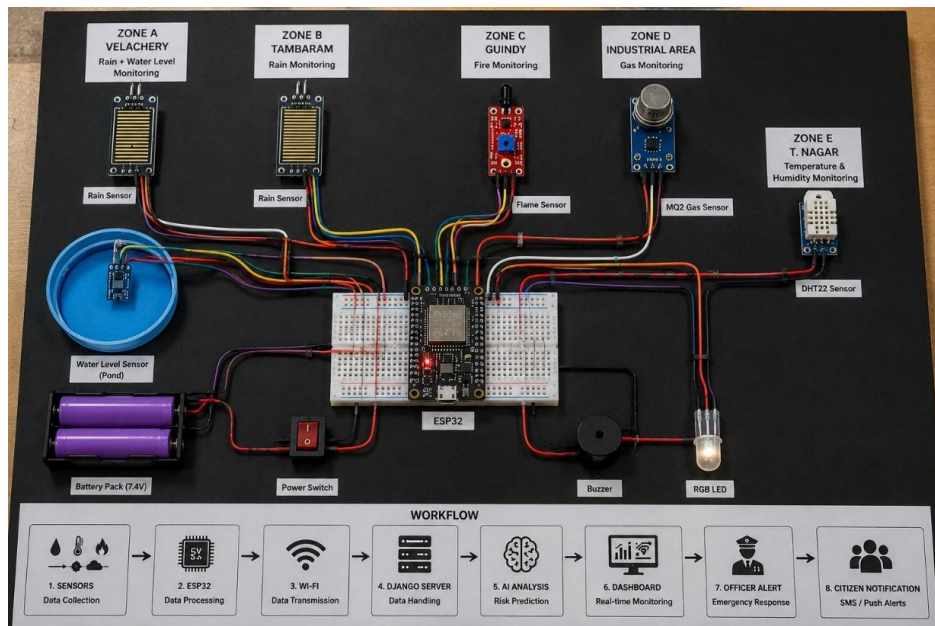


Figure A.2 Physical prototype evidence from the implemented SentinelAI project.

The architecture evidence shows the major data acquisition, data processing, AI disaster detection, intelligent decision, monitoring/analysis and emergency-response stages.

Appendix B – Core API / Interface Summary

Table 14. Core API / Interface Summary

Endpoint / Interface	Purpose
POST /api/v1/sensor-data/	Receive IoT sensor data.
GET /api/v1/zones/	Retrieve monitored zones.
GET /api/v1/sensors/	Retrieve sensor/device status.
GET /api/v1/predictions/	Retrieve risk predictions.
POST /api/v1/incidents/	Create an incident.
PATCH /api/v1/incidents/{id}/	Update incident state.
GET /api/v1/officers/	Retrieve officer availability.
POST /api/v1/assignments/	Assign/recommend responder.
POST /api/v1/citizen-reports/	Create citizen report.
GET /api/v1/reports/{token}/	Track public report status.
POST /api/v1/routes/	Calculate response/evacuation route.
GET /api/v1/analytics/	Retrieve operational analytics.

Appendix C – Incident and Response Logic

The operational chain is: detect → validate → assess risk → create/prioritize incident → identify eligible responder → assign/recommend → start response timer → accept/in-progress → route → resolve → close → record audit/performance information.

The responder recommendation considers the incident zone, officer availability, skill compatibility, distance or last known operational location, workload and configured performance factors. The project working record gives an example suitability model of 40% availability, 25% distance, 20% skill match, 10% workload and 5% historical performance; these weights are prototype configuration rather than an official emergency-service policy.

Appendix D – Hardware Components

Table 15. Hardware Components

Component	Purpose / Role
ESP32 DevKit V1	Central controller and Wi-Fi data transmission
Rain Sensor	Rainfall condition detection
Water Level Sensor	Flood/water accumulation demonstration
Flame Sensor	Fire condition demonstration
MQ-2 Gas Sensor	Gas-leak condition demonstration
DHT11/DHT22-class Sensor	Temperature and humidity observation
Buzzer	Local alert indication
RGB LED	Local state indication
Breadboard + jumper wires	Prototype interconnection
Battery pack / USB power	Prototype power source

Appendix E – Experimental / Raw Data Record Structure

The prototype stores sensor observations with device/zone context, value and timestamp. For a production evaluation, each experiment should additionally record the configured threshold, observed input, risk state, incident state, assignment timestamp, acceptance timestamp, route calculation time and final resolution time. This structure supports later quantitative evaluation without fabricating values in the capstone report.

Appendix F – Individual Contribution Evidence

Table 16 . Individual Contribution Evidence

Evidence Item	What It Demonstrates	Individual Relevance
System architecture figure	Layered integration of sensing, data, intelligence, geospatial and response modules.	Supports architecture and integration contribution.
Physical prototype photograph	Actual ESP32 and sensor integration.	Supports physical-to-digital integration and demonstration work.
API/interface design	Defined boundaries between sensing, backend and response services.	Supports system integration responsibility.
End-to-end test matrix	Verification across sensor, risk, incident, responder, route, citizen and security workflows.	Supports testing and engineering judgment.
Deployment/build verification records	Evidence that the software was packaged and validated for demonstration.	Supports integration and deployment responsibility.
Individual reflection	Personal engineering decisions, learning and limitations.	Supports independent learning outcome.

Appendix G – Capstone Project Outcome Mapping

Table 15. Capstone Project Outcome Mapping

Outcome / Area	Evidence in This Individual Report
Complex engineering problem formulation	Chapter 1 problem statement and constraints
Engineering design under constraints	Chapter 3 requirements, alternatives, trade-offs and risk register
Written/oral technical communication	Whole report, tables, figures and viva preparation
Ethics and professional responsibility	Chapter 3.7 and declaration
Teamwork and leadership	Chapter 4.8 roles and integration responsibilities
Experimentation and engineering judgment	Chapter 4.3–4.5 test plan and analysis
Independent acquisition of knowledge	Chapter 5.3 reflection
Standards	Chapter 3.2
Sustainability and societal impact	Chapter 3.7
Risk assessment and mitigation	Chapter 3.7
Security considerations	Chapter 3.7 and Appendix F
Integrated/system-level engineering	Chapter 3.5 and Figures 3.1/A.1
Project planning and management	Chapter 4.8
Individual contribution	Contribution Statement, Chapter 4 and Appendix F

